

Phase B — Fake-league characterization (not curve fitting)

Knobs (mini glossary)

Symbol	Plain name	What it controls
A_i	Player ability	Innate talent draw (Beta(2,2) on [0,1]) 539 Preset
T_j	Team talent target	Synthetic i.i.d. Uniform[0,1] per team— ideal “seat” for soft assignment
ρ	Assignment assortativity	How tightly players match team talent targets
L_c	Congestion measure	LOO peer viability Smooth - mean of $\sigma(\gamma(A_j - \theta))$ over teammates Hard option in code ($A_j > \theta$)
λ	Congestion weight in score	How hard L_c penalizes ranking – not congestion itself
θ	Viability cutline	Center of the sigmoid
γ	Sigmoid sharpness	Steepness of viability step around θ
K/N	Selectivity	Fraction of league selected ($K \div N$)

Not curve fitting: which rules **bend** synthetic selection curves — not NCAA fit yet.

- Three steps: ASSIGN (ρ) $\rightarrow$ SCORE ($S_i = A_i - \lambda L_c$) $\rightarrow$ SELECT ($top - K$)
- then VISUALIZE by pool-mean bin.
- L_c in this deck: `\texttt{crowding\_smooth}` (mean σ). Code also builds hard L_c = LOO share with $A_j > \theta$ — not shown here.
- One-at-a-time (OAT): each slide moves one knob; other settings stay at benchmark values.
- Seed 42 on stochastic steps: draw A_i, T_j ; soft- ρ assignment.
- Score, top- K , bins are deterministic.
- League: $N = 5600, K = 560, K/N = 10\%$ (539 preset).
- **A**
- $\lambda = 0.55$ — **tier1_539_reference_settings.json / playground default**
- $\theta = 0.72, \gamma = 10$ — same 539 reference JSON
- $\rho = 8$ — gallery fixed assign for λ/θ slides (moderate-high mixing; not a 539 JSON field)
- $K/N = 10\%$ — characterization default (MBB draft $\approx 1\%$ is a separate calibration point)

Real hero: draft rate vs **poolq_loo** (teammate quality, leave-one-out) — a **bend** at the top bins.

- This deck: fraction selected vs **pool mean** in a fake league — same *kind* of question, different axis and league.
- Benchmarks anchor to the Alex 539 playground track so sim slides stay comparable
- We have **not** re-fit those numbers to NCAA data in Phase B.

Claim: Phase B maps which generative knobs move **who gets selected** — prerequisite before Phase C calibration to the hero.

Phase B — Characterize ρ (assignment assortativity) — sort-and-chop suppressed

Characterize ρ — one-at-a-time at 539 baseline (score + top- K fixed)

ASSIGN (knob ρ) — LaTeX:

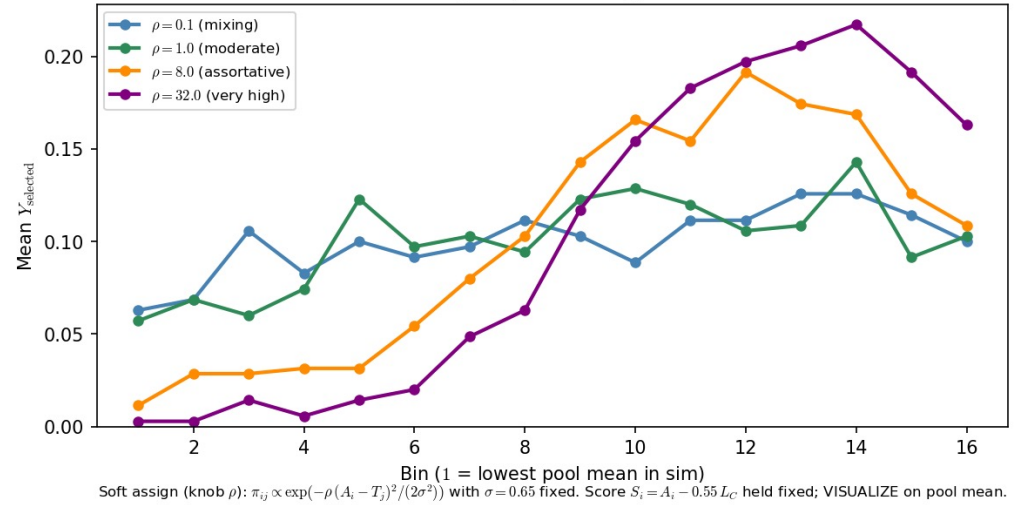
$$\pi_{ij} \propto \exp\left(-\rho \cdot \frac{(A_i - T_j)^2}{2\sigma^2}\right)$$

- $\sigma = 0.65$ fixed
- arms: $\rho \in \{0.1, 1, 8, 32\}$
- $S_i = A_i - 0.55 \cdot L_C$ (crowding in score held constant across arms)
- top- K by S_i
- VISUALIZE = mean Y_{selected} vs pool mean (16 bins)

Pass C — assignment ablation (539, score + top- K fixed)

$S_i = A_i - 0.55 L_C \mid N = 5600, K = 560, K/N = 0.10 \mid 16 \text{ bins}$

(soft ρ arms only)



OAT: one draw of A_i, T_j (seed 42); only ASSIGN changes across arms.

Low ρ : players mix across teams — selection vs pool mean stays nearly flat.

High ρ : assortative matching — peer environments concentrate; inverted- U emerges.

Sort-and-chop arm omitted from plot (still in CSV bundle; toggle restores it).

Claim: at fixed score and top- K , ρ in assignment matters — low- ρ mixing yields a nearly flat curve; high- ρ assortativity delivers the peer-pressure inverted- U readout.

Phase B — Characterize λ (congestion in score)

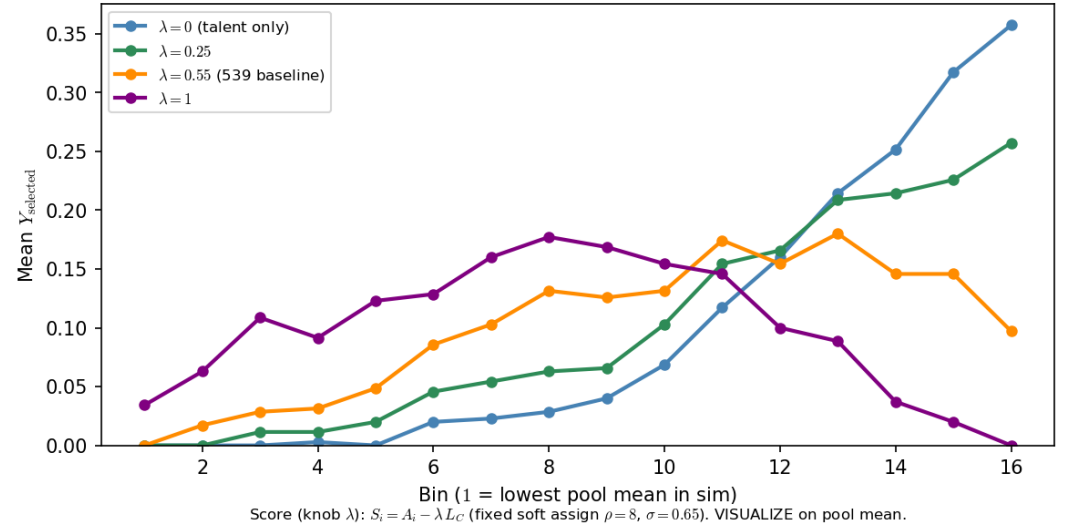
Characterize λ — one-at-a-time at 539 baseline (assign + top- K fixed)

$$S_i = A_i - \lambda \cdot L_c$$

- $\rho = 8$ fixed (soft assign; same roster all arms)
- $\sigma = 0.65$ fixed
- arms: $\lambda \in \{0, 0.25, 0.55, 1\}$
- top- K by S_i
- VISUALIZE = mean $Y_{selected}$ vs pool mean (16 bins)

Pass B — score ablation (539, assign + top- K fixed)

$\rho = 8$ fixed | $N = 5600$, $K = 560$, $K/N = 0.10$ | 16 bins



OAT: one draw of A_i, T_j + one assignment (seed 42); only SCORE changes.

$\lambda = 0$: $S_i = A_i$ only — selection vs pool mean stays roughly monotone.

$\lambda > 0$: congestion in score — inverted- U / roster-pressure readout emerges.

Pairs with Pass C: λ bends score; ρ shapes the sorted roster world.

Claim: at fixed ρ and top- K , λ in score matters — $\lambda = 0$ yields a roughly monotone talent ladder; $\lambda > 0$ puts roster pressure in the advancement rule and bends the curve.

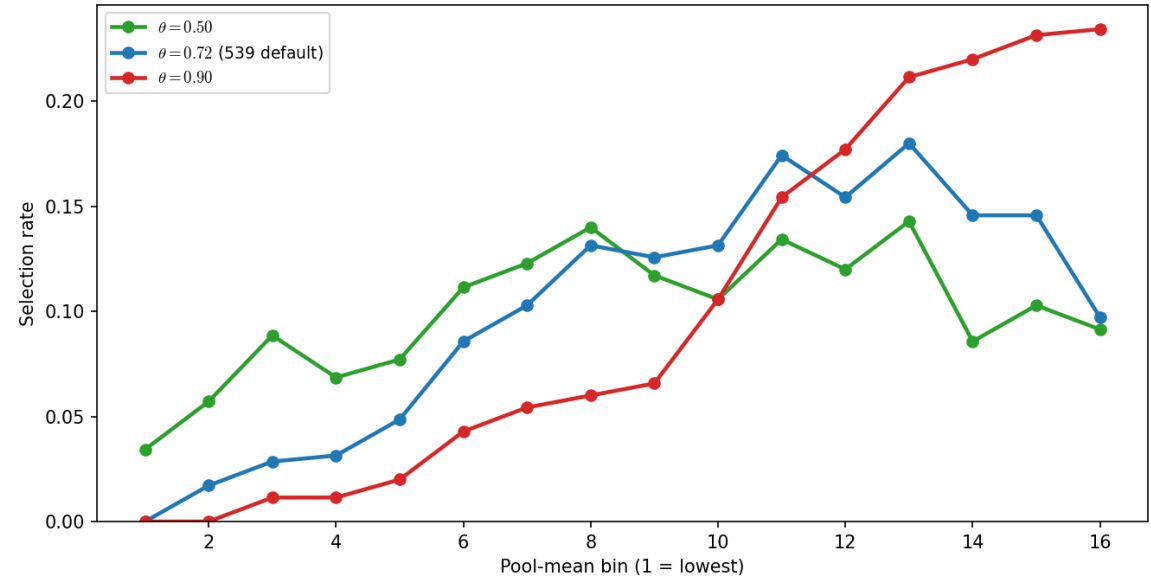
Phase B — Characterize θ (viability cutline) — OAT at $K/N = 10\%$

Characterize θ — one-at-a-time at 539 baseline (assign + score + top- K fixed)

$$L_c = \sigma(\gamma(A_i - \theta))$$

- $\rho = 8, \lambda = 0.55, \gamma = 10$ fixed
- arms: $\theta \in \{0.50, 0.72, 0.90\}$
- $K/N = 10\%$ ($K = 560, N = 5600$)
- VISUALIZE = mean $Y_{selected}$ vs pool mean (16 bins)

θ OAT at $K/N = 0.10$ (539) — soft $\rho = 8, \lambda = 0.55, \gamma = 10$
 $N = 5600, K = 560, K/N = 0.10$ | 16 bins | seed=42



OAT: soft $\rho = 8, \lambda = 0.55, \gamma = 10$ fixed; only θ in $\sigma(\gamma(A - \theta))$ changes.

$\theta = 0.50$: mid-pool hump (peak bin ≈ 13).

$\theta = 0.72$ (539 default): similar hump, slightly higher peak rate.

$\theta = 0.90$: top bins dominate — curve becomes monotone-increasing at the elite edge.

Claim: at fixed ρ, λ , and $K/N = 10\%$, θ (viability cutline) moves the readout — low θ keeps a mid-pool hump; high θ top-saturates the curve.

Phase B — Characterize θ — panel with K/N (selectivity)

Characterize θ — one-at-a-time at 539 baseline (K/N varies; assign + score rule fixed)

$$L_c = \text{mean}_{j \neq i} \sigma(\gamma(A_j - \theta))$$

- $\rho = 8, \lambda = 0.55, \gamma = 10$ fixed
- $\theta \in \{0.50, 0.72, 0.90\}$
- $K/N \in \{1\%, 10\%, 40\%\}$ — heatmap = peak pool-mean bin
- Readout: where the inverted- U peak sits vs selectivity

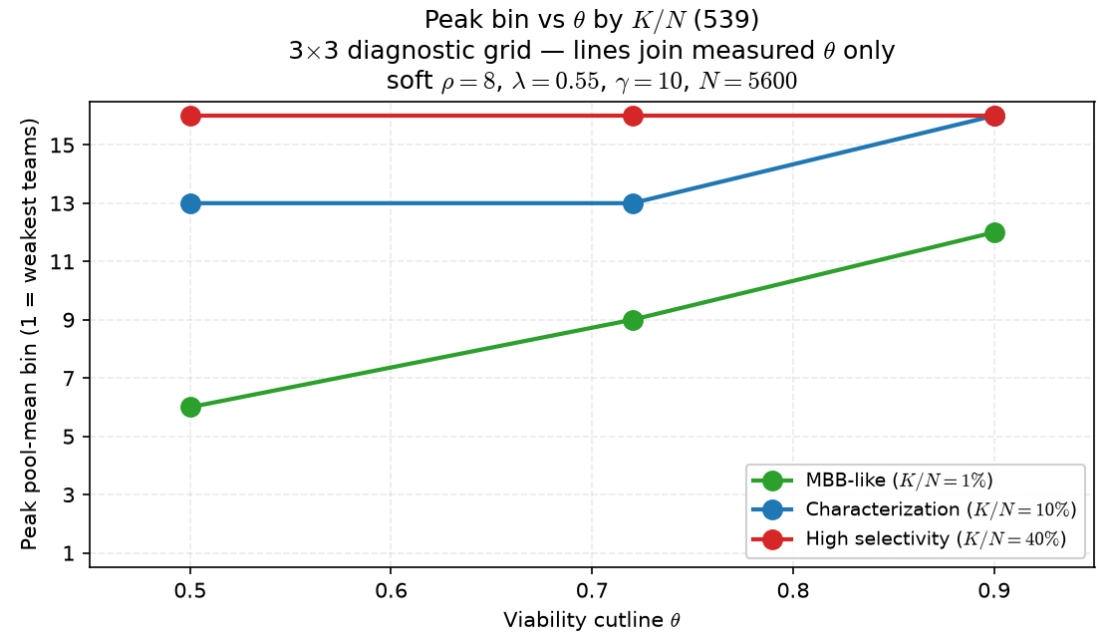
Grid: $\theta \in \{0.50, 0.72, 0.90\} \times K/N \in \{1\%, 10\%, 40\%\}$ on $N=5600$.

At $K/N = 1\%$: peak bin rises 6 $\rightarrow$ 9 $\rightarrow$ 12 as θ increases.

At $K/N = 10\%$: $\theta = 0.90$ pushes peak to bin 16 (top saturation).

At $K/N = 40\%$: all arms peak at bin 16 — selectivity dominates θ .

Do not fix $\theta = f(K/N)$ without Alex — panel is descriptive.



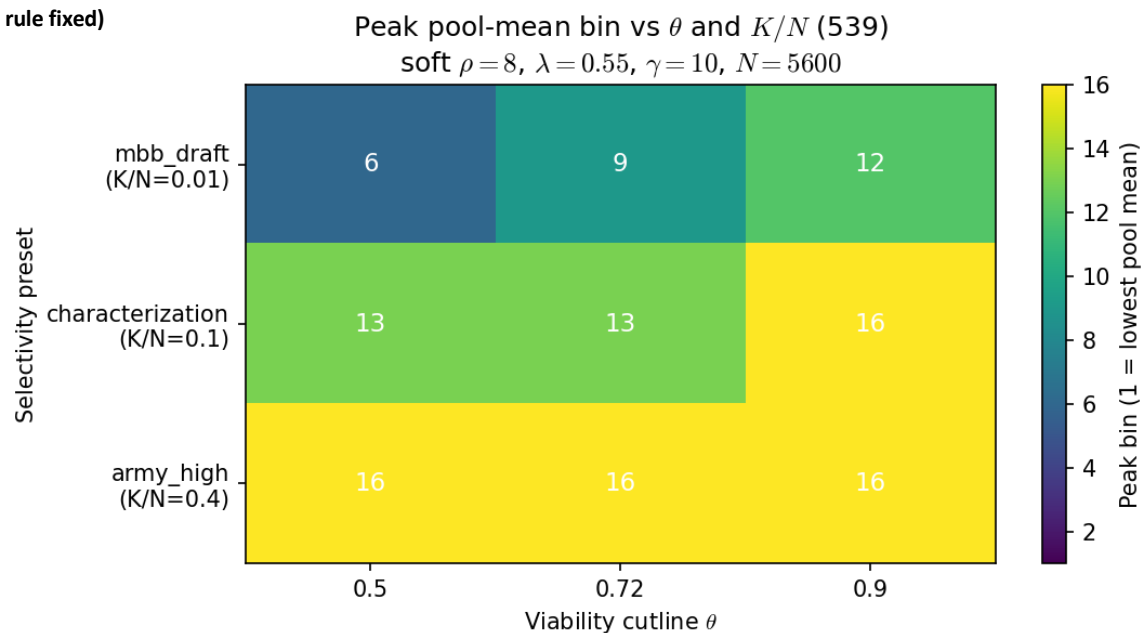
Claim: θ and K/N co-vary — at MBB-like selectivity (1%) raising θ shifts peak bin up the pool ladder; at 40% selectivity curves are top-saturated regardless of θ .

Phase B — Characterize θ — panel with K/N (selectivity)

Characterize θ — one-at-a-time at 539 baseline (K/N varies; assign + score rule fixed)

$$L_c = \text{mean}_{j \neq i} \sigma(\gamma(A_j - \theta))$$

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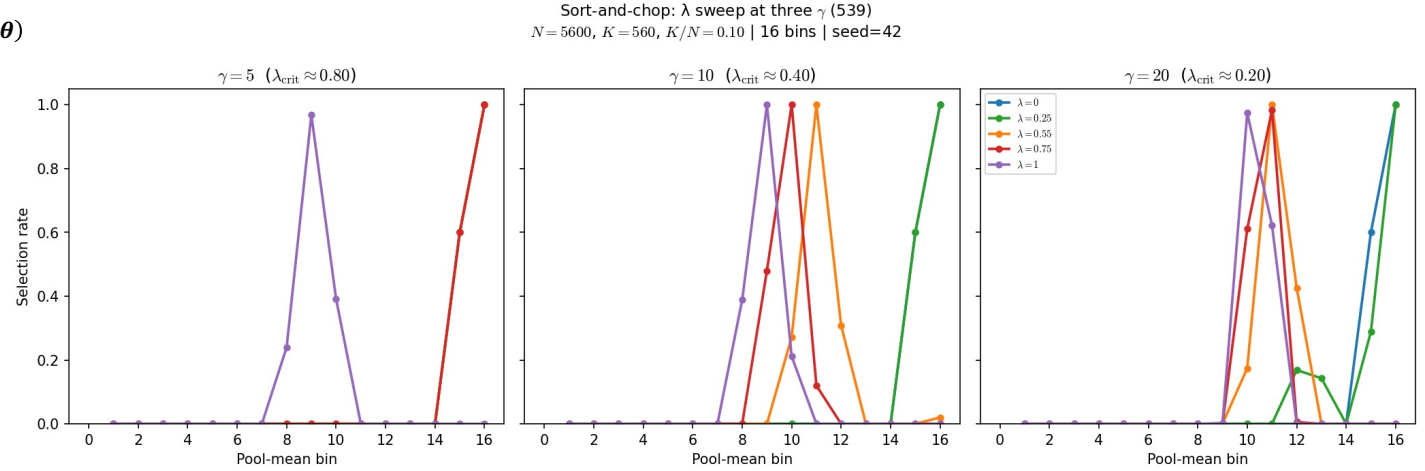
Claim: θ and K/N co-vary — at MBB-like selectivity (1%) raising θ shifts peak bin up the pool ladder; at 40% selectivity curves are top-saturated regardless of θ .

Phase B — Characterize γ (viability sharpness) — selection curves

Characterize γ — one-at-a-time (539 sort-and-chop benchmark)

$$\sigma(x) = \frac{1}{1 + e^{-x}}, x = \gamma(A - \theta)$$

- $\theta = 0.72$ fixed; sort-and-chop assign
- arms: $\gamma \in \{5, 10, 20\}$
- within each γ : $\lambda \in \{0, 0.25, 0.55, 0.75, 1.0\}$
- VISUALIZE = mean $Y_{selected}$ vs pool mean (16 bins)



Sort-and-chop benchmark: same rosters per γ ; only γ and λ in score change.

$\gamma = 5$: $\lambda_{crit} \approx 0.8$ — hump needs larger λ .

$\gamma = 10$ (539): $\lambda_{crit} \approx 0.4$ — curve morph near $\lambda \gtrsim 0.4$.

$\gamma = 20$: $\lambda_{crit} \approx 0.2$ — congestion reorders at modest λ .

Below crit: monotone elite-edge shape; above crit: inverted- U /hump.

Claim: γ (sigmoid slope in L_C) shifts when congestion bites — higher γ lowers $\lambda_{crit} \approx 4/\gamma$ on sort-and-chop; below crit, λ arms can look identical.

Phase B — $\lambda_{crit} \approx 4/\gamma$ (sort-and-chop algebra)

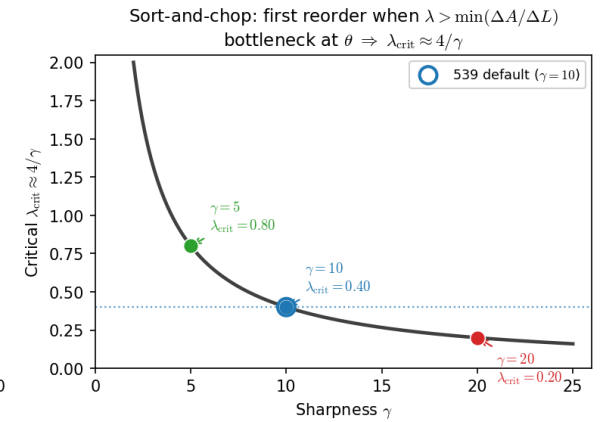
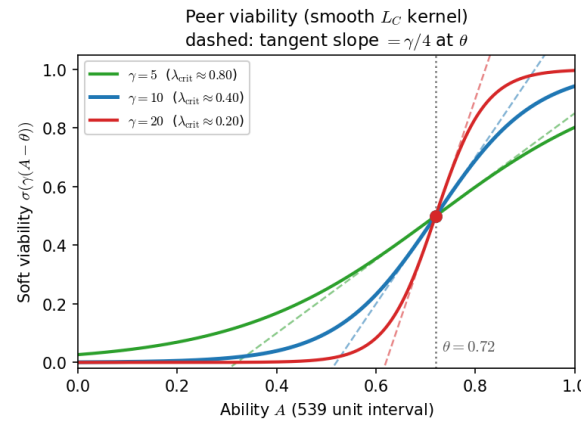
Characterize γ — one-at-a-time (539 sort-and-chop benchmark)

$$\sigma(x) = \frac{1}{1 + e^{-x}}, x = \gamma(A - \theta)$$

- Educational: how γ shapes L_c along ability axis
- vertical lines: $\lambda_{crit} = 4/\gamma$ for each γ
- 539 defaults: $\theta = 0.72, \gamma = 10 \rightarrow crit = 0.4$
- Pairs with sort-and-chop λ threshold memo (doc 06)

Why $\lambda \approx 0.41$ on sort-and-chop (not A_i , not T_j)

$S_i = A_i - \lambda L_C$; γ only matters when $\lambda > 0$ (crowding in score)



Soft viability: $\sigma(\gamma(A - \theta))$ — not a hard step; γ sets knee steepness.

On sort-and-chop, $A_i \parallel L_C$ within teams $\rightarrow$ small λ may not reorder top- K .

λ_{crit} : first λ where $S_i = A_i - \lambda L_C$ changes who gets selected.

Visual curve departure in 16 bins can lag λ_{crit} — use finer bins or check reorder.

Ties to Alex Q: why $\lambda = 0$ and $\lambda = 0.25$ match until $\lambda > 4/\gamma$.

Claim: $\lambda_{crit} \approx 4/\gamma$ is the first λ where score reorder beats talent-only on sort-and-chop — not when the curve visually departs from $\lambda = 0$ in coarse bins.